SAM OBILA ALLELA

MACHINE LEARNING ENGINEER | ROBOTICS & APPLIED AI
Nairobi, Kenya

LinkedIn
Substack
GitHub
Medium

PROFESSIONAL SUMMARY

Machine learning engineer working at the intersection of artificial intelligence, robotics and inclusive technology. Translates AI capabilities into low-latency human-robot interaction, working across robotics hardware, inference pipelines, and classroom demonstrations with Deaf learners. I contribute to assistive technology shaped for African learning contexts and inclusive STEM education.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer, Zerobionic Africa | Jun 2026 – Present

Nairobi, Kenya · Hybrid

- Spearhead development of machine-learning capabilities behind Zerobionic's robotic platforms, working across speech intelligence, model integration and real-time human-robot interaction.
- Build and optimize speech-to-text and text-to-speech pipelines that allow robotic systems to listen, interpret spoken input and support accessible interaction.
- Work with robotics and hardware engineers to integrate ML inference into physical systems, moving models from experimentation into low-latency robotic behavior across a platform deployed in 78 schools..
- Participate in school outreach and demonstrations with Deaf learners, using direct classroom interaction to understand accessibility needs and connect engineering decisions to the people the technology is intended to serve.
- Contribute to the broader development of assistive robotics built in Africa for inclusive STEM education and communication.
- Contribute to a platform now deployed in 78 schools across Kenya, with a reported accuracy rate of 92%.

Independent AI/ML Engineer, Freelance / Independent | 2024 – Present

Remote

- Design and build applied AI systems for product and engineering use cases, from early problem framing and prototyping through model integration and deployable software.
- Developed retrieval and knowledge systems using RAG, vector search and domain-specific model adaptation for document-intensive workflows.
- Built autonomous and agentic systems spanning AI observability, security analysis and mobile workflows, with emphasis on reliable orchestration, human oversight and production usability.
- Work across Python and TypeScript ecosystems, combining machine-learning components with APIs, cloud services and user-facing applications.

RESEARCH & PUBLICATION

Research Contributor / Co-author, S Labs | 2026

Alternative Credit Infrastructure for the Informal Economy, Israel Gilbert & Sam Obila, SSRN, July 2026.

- Co-authored a 29-page research review examining alternative credit infrastructure for informal economies in Kenya, Uganda and Ghana, with attention to privacy, regulation and responsible financial AI.
- The work examines technical and institutional constraints around machine-learning-based credit systems and frames an execution roadmap for privacy-preserving financial infrastructure. Read paper on SSRN

COMMUNITY LEADERSHIP & SERVICE

STEM Tutor & Youth Mentor (Volunteer), Bunifu Youths Kenya | Current

- Tutor and mentor young learners in practical technology skills, supporting hands-on learning in coding, robotics, AI and problem-solving.
- Help learners move from consuming technology to building with it by guiding projects, explaining technical concepts in accessible language and encouraging experimentation.
- Support a youth-development community whose programs include school outreach, coding clubs, robotics activities, bootcamps and mentorship for children and teenagers across Kenya.

Community & Event Volunteer, GDG Nairobi | Current

- Volunteer with Nairobi's developer community, supporting community learning and technology events that connect developers, students and emerging builders including DevFest Nairobi, attended by 1,500+ developers as of 2023.
- Contribute to an ecosystem centered on accessible technical learning through workshops, hackathons, study sessions and developer events spanning AI, cloud, web and related technologies, including a Cloud session with 195 registrants as of 2024.
- Support community participation and peer learning in a volunteer-led network focused on strengthening Kenya's developer ecosystem.

Accessibility & STEM Outreach, Zerobionic Africa | 2026 – Present

- Take part in outreach to schools serving Deaf learners, demonstrating assistive robotics and engaging students around how AI and robotics can support access to STEM learning in a country where only about 20,000 of an estimated 300,000 children with hearing impairment were enrolled in school as of 2026.
- Use feedback and observations from learners and educators to better understand the practical realities of inclusive technology deployment beyond the engineering lab.

TECHNICAL WRITING & KNOWLEDGE SHARING

I publish accessible writing on AI engineering, model evaluation and emerging AI infrastructure through Substack and Medium. Selected work:

The Universal Port for AI: A Deep Dive into MCP Architecture | May 2026

How Do You Know Your AI Is Actually Good? A Guide to LLM Evaluation | Mar 2026

Teaching AI to Remember: A Deep Dive into Retrieval-Augmented Generation | Mar 2026

Substack: @allela | Medium: @obilasam3

EDUCATION

Bachelor of Science in Mathematics & Computer Science, Multimedia University of Kenya | 2026
Nairobi, Kenya
Foundation in algorithms, linear algebra, probability and statistics, distributed systems and computational problem-solving.

PROFESSIONAL DEVELOPMENT

- ✓ **AWS AI Practitioner Challenge**, Udacity, 2026
- ✓ **AI Engineer for Data Scientists Associate**, DataCamp, 2026
- ✓ **Google AI Professional Certificate**, Google Career Certificates, 2026
- ✓ **Python Data Associate**, DataCamp, 2026

SKILLS

Technical Skills

- Python
- TypeScript
- FastAPI
- React/Next.js • Docker
- Elasticsearch
- Vector databases
- Cloud deployment

Applied AI Systems

- Model deployment
- Machine learning pipelines
- Speech recognition
- Speech synthesis
- MLOps

Robotics AI

- Human-robot interaction
- Model integration
- Agentic systems

- RAG
- LLM evaluation